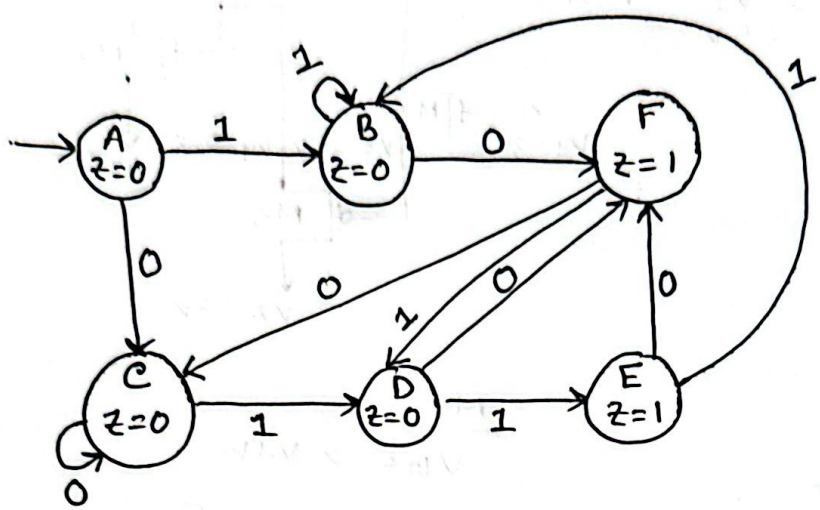
Azmari Sultana

Summer 2025 (A)

1

a) Moore type FSM

b) 10 or 011 overlapping



c)

Present state $y_3y_2y_1$	Next state $Y_3Y_2Y_1$		Output, z
	w=0	w=1	
A (000)	C (010)	B (001)	0
B (001)	F (101)	B (001)	0
C (010)	C (010)	D (011)	0
D (011)	F (101)	E (100)	0
E (100)	F (101)	B (001)	1
F (101)	C (010)	D (011)	1
(110)	d	d	d
(111)	d	d	d

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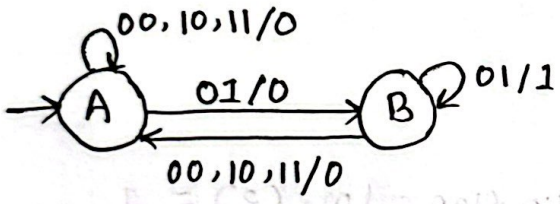
d) $\underline{z}$

$y_3 y_2$ \ y_1	0	1
00	0	0
01	0	0
11	d	d
10	1	1

$z = y_3$

(2)

a) Mealy type FSM



b)

Present state y_1	Next state Y_1				Output			
	00	01	10	11	00	01	10	11
A (0)	A (0)	B (1)	A (0)	A (0)	0	0	0	0
B (1)	A (0)	B (1)	A (0)	A (0)	0	1	0	0

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c) Y_1

y_1 \ xy	00	01	11	10
0	0	$\boxed{1}$	0	0
1	0	$\boxed{1}$	0	0

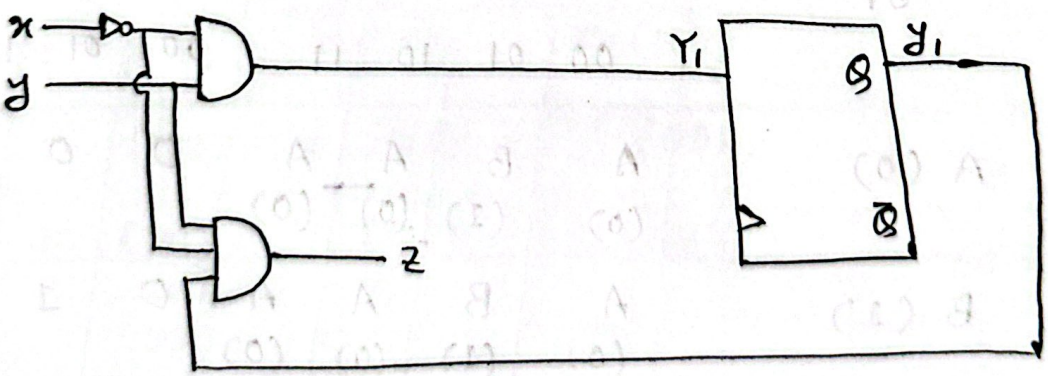
$Y_1 = \bar{x}y$

z

y_1 \ xy	00	01	11	10
0	0	0	0	0
1	0	$\boxed{1}$	0	0

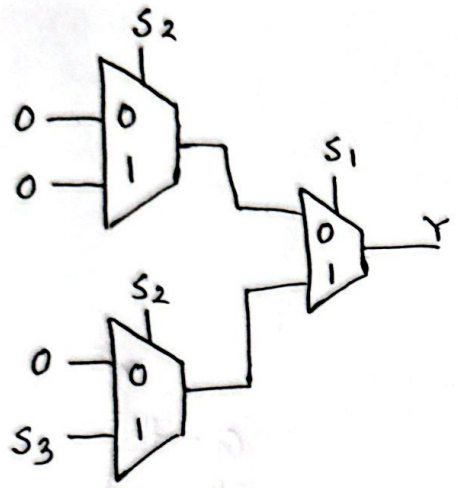
$z = y_1\bar{x}y$

d) Number of flip flop $= \log_2(2) = 1$



③

a) $Y = S_1 \cdot S_2 \cdot S_3$



S_1	S_2	S_3	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

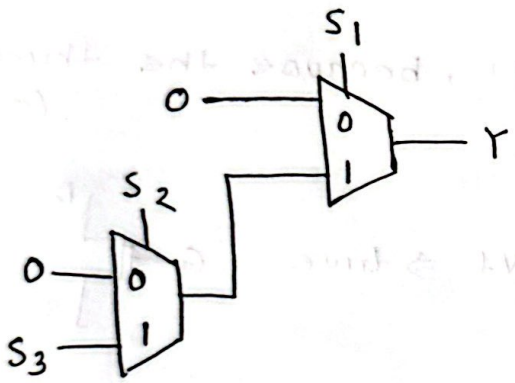


Fig: 2:1 MUX

b) Figure 1 is a negative edge triggered D Flip Flop. Because the first latch is transparent when clk is high and the second latch is transparent when clk is low.

Device 1 is a positive latch.

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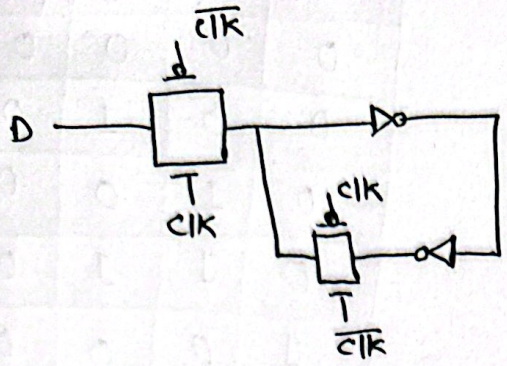
c)

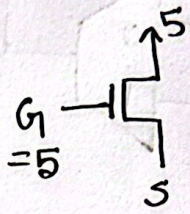


d) NMOS

$V_{GS} > V_t$

$\Rightarrow 5 - V_S > 0.1$

$\Rightarrow V_S \leq 4.9$

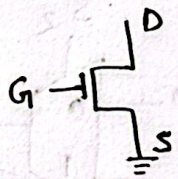


So NMOS pass 'bad 1', because the threshold is 4.9 less than 5.

$V_{GS} > V_t$

$V_{GS} = 5 - 0 = 5 > V_t \Rightarrow$ true

pass 'good 0'

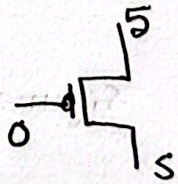


PMOS

$V_{SG} > |V_{tP}|$

$\Rightarrow V_S - 0 > 0.1 \Rightarrow$ true

~~$\Rightarrow$~~ So, pass 'good 1'



when it pass 0, it increase it.

$V_{SG} = V_{out} - 0 = V_{out}$

$V_{SG} = |V_{TP}| = V_{OUT}$

Bad '0'

Azmani

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